# Environmental impact and optimisation potential of the construction phase of seasonal underground thermal energy storage systems

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## ABSTRACT

Underground thermal energy storage (UTES) systems buffer fluctuations between the heating and cooling surpluses and demands in modern energy systems. They allow for the integration of intermittent renewable energy sources while reducing the consumption of conventional energy. In particular, seasonal storage offers the opportunity to manage heat and cold supply more sustainably and to reduce greenhouse gas (GHG) emissions. So far, numerous seasonal thermal energy storage systems have been realised, and several technology variants have achieved appropriate market maturity. However, no comprehensive environmental assessment is available yet that analyses the different solutions regarding their life cycle phases. There, the construction phase deserves foremost attention, as it may differ substantially depending on storage type, individual devices implemented, and size of an individual facility.

Our study analyses the environmental impacts of the construction phase of three different UTES types using the life cycle assessment (LCA) approach according to ISO standards 14040 and 14044. This includes a tank thermal energy storage (TTES) operated in the city of Munich (Germany), a water-gravel thermal energy storage (WGTES) in Eggenstein-Leopoldshafen (Germany), and a pit thermal energy storage (PTES) in Marstal (Denmark). The results are contrasted with those of an aquifer thermal energy storage (ATES) in Bonn (Germany).

The results of the LCA identify and quantify different influencing factors on the environmental impacts of the construction phase of these sites, highlighting key differences and potential for improvements. For example, environmental benefits can be achieved by using water as a filling material, employing underground constructions, and using large storage volumes with small surface-to-volume ratio. In contrast, the use of materials such as concrete, steel, foam glass gravel, or polyethylene represents an environmental burden that should be avoided, e.g., by substitution with environmentally friendly alternatives without compromising the storage capacity.

Such findings from environmental assessments are crucial to optimise technological choice and design. They offer quantitative environmental arguments to evaluate potential environmental benefits. In the long term, a complete cradle-to-grave assessment will help to highlight the potential advantages of a broader deployment of UTES.

## 1. INTRODUCTION

Space heating and domestic water demand account for a significant share of total energy consumption and greenhouse gas (GHG) emissions (Bayer et al., 2012), with most of the energy still supplied by fossil fuels (Bott et al., 2019). Underground thermal energy storage (UTES) systems have the potential to reduce these impacts (Pinel et al., 2011; Xu et al., 2014). Life cycle assessment (LCA) represents the standardised framework for evaluation of product- or service-related impacts. While LCA is increasingly applied to seasonal thermal energy storage (sTES), most studies focus on their operational phase, particularly for aquifer thermal energy storage (ATES) (Bloemendal et al., 2014; Fleuchaus et al., 2018; Schüppler et al., 2019), where electricity consumption for heat pumps dominates environmental impacts (Ni et al., 2020; Tomasetta et al., 2015). This study analyses the environmental impact of the main materials used during the construction phase of three representative UTES case studies. Results are evaluated against an ATES in Bonn (Germany), aiming to identify optimisation potentials for sustainable storage materials. A more detailed analysis of all materials and processes, as well as transportation and a comparison with further ATES systems, is presented by Weise et al. (2025).

## 2. APPLICATION CASES

As demonstrated in Table 1, these systems vary significantly in terms of site-specific conditions, storage volumes/capacity, and construction materials. Consequently, a direct one-to-one comparison is not appropriate. Nevertheless, common trends and key differences can be identified by using LCA.

**Table 1: Key parameters of three selected sTES facilities: tank thermal energy storage (TTES) in Munich-Ackermannbogen (Germany), water-gravel thermal energy storage (WGTES) in Eggenstein-Leopoldshafen (Germany), and pit thermal energy storage (PTES) in Marstal (Denmark), A/V ratio: surface-to-volume ratio, WEV: water equivalent volume.**

| Parameter | TTES | WGTES | PTES |
|---|---|---|---|
| A/V ratio | 0.287 | 0.428[1] | 0.233[1] |
| Volume (m³) | 6,000[2,3] | 4,500[4,5] | 75,000[5,6] |
| WEV (m³) | 5,700 [4,7,8] | 3,000[9] | 75,000[10] |
| Capacity (MWh) | 480[2] | n.a. | 6,960[6,10] |
| Energy input | 969[2,9] | 287[9] | 7,538[11] |
| Energy output | 809[2,9] | 226[9] | 4,141[11] |
| Lifetime expected (a) | 30 | 30 | 30 |

[1] (Xiang et al., 2022), [2] (Reuß, 2016), [3] (Spliethoff and Stimming, 2010), [4] (Guadalfajara et al., 2014), [5] (Xiang et al., 2022), [6] (Holm and PlanEnergi, 2021), [7] (Röpke, 2007), [8] (Keil et al., 2008), [9] (Mangold et al., 2012), [10] (Jensen and PlanEnergi, 2014), [11] (Mangold et al., 2016).

Since 2007, a 6,000 m³ tank thermal energy storage (TTES) in Munich (Germany) has supplied heat to 320 residential units (Spliethoff and Stimming, 2010). This partially buried sTES uses steel and concrete reinforcements for stability (Figure 1a) (Mangold et al., 2007; Spliethoff and Stimming, 2010). Due to high costs of gravel and large amounts of excavated soil, a water-filled technology variant was chosen (Mangold et al., 2007). The excavated soil was reused for covering (Spliethoff and Stimming, 2010).

In Marstal (Denmark) a 75,000 m³ pit thermal energy storage (PTES) was built in 2012 to store solar thermal energy for a district heating system (Dannemand Andersen et al., 2013). This facility is completely embedded within the ground, eliminating the need for steel or concrete support. Instead, plastic foils are employed for lining, and the excavated soil was reused to form an embankment around the storage and to increase its volume (Figure 1b) (Dannemand Andersen et al., 2013; Sørensen and Schmidt, 2018).

Third, in Eggenstein-Leopoldshafen (Germany), a water-gravel thermal energy storage (WGTES) was built for integration into a renovated school complex (Bodmann et al., 2005; Marx et al., 2012; Ochs et al., 2008). To maintain the usability of the surface above as a schoolyard, the storage was filled with (partly excavated) sand and gravel. Similar to the PTES, the use of static supporting material is not required (Figure 1c) (Mangold et al., 2007; Schneider, 2009).

## 3. LIFE CYCLE ASSESSMENT

This study follows the standardised procedure of DIN EN ISO 14040 (2021) and DIN EN ISO 14044 (2018) to quantify construction phase environmental impacts for the three sTES described above. It focuses on global warming, applying the impact assessment method *IMPACT2002+* (Jolliet et al., 2003) in *SimaPro 10.0.0.28* with data from the database *ecoinvent 3.10*. System boundaries include sealing (high-density polyethylene (HDPE), polypropylene (PP)), insulation (foam glass gravel (FGG), expanded glass granulate (EGG)), and structural elements (steel, concrete). Further materials used are summarised in the category 'Others' (Table 2).

**Table 2: Life cycle inventory of sealing, insulation and static elements of the three sTES, HDPE: high-density polyethylene, EGG: expanded glass granulate, FGG: foam glass gravel.**

| Components | TTES in t | WGTES in t | PTES in t |
|---|---|---|---|
| HDPE | 0.9 | 3.6 | 113 |
| Polypropylene | - | 2.3 | - |
| EGG | 149 | 102 | - |
| Polymer foam | - | - | 72.7 |
| FGG | 16.7 | 101 | - |
| Steel | 27 | - | - |
| Concrete | 898 | - | - |

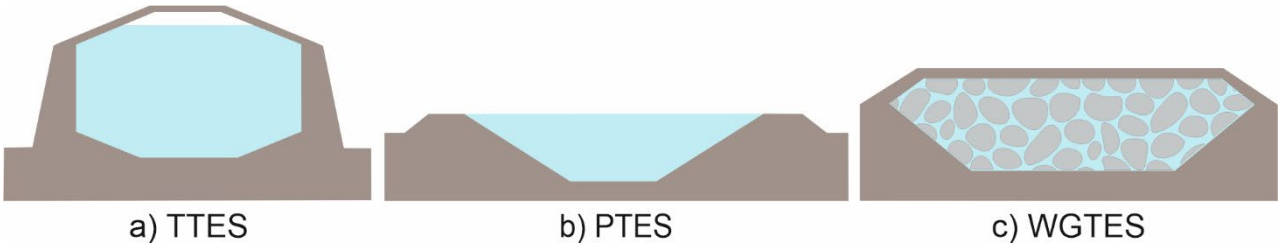


**Figure 1: Schematic representations of the three selected sTES systems: a) tank thermal energy storage (TTES), b) pit thermal energy storage (PTES), and c) water-gravel thermal energy storage (WGTES); modified to (Weise et al., 2025).**

This category involves additional construction materials not included in the main categories, primarily small amounts of steel as well as other materials with relatively low global warming impacts, such as wood, soil, gravel, and polytetrafluoroethylene (PTFE). The functional unit is defined as 1 $kWh_{th}$ of thermal energy output over one reference year, with an expected lifetime of 30 years. The goal of the LCA is to identify and evaluate environmental impacts under site-specific conditions, focusing on main construction materials.

## 4. RESULTS AND DISCUSSION

### 4.1 IMPACT ASSESSMENT OF MATERIAL

In this section, the environmental impacts of materials of the sTES systems are analysed (Figure 2). We focus on global warming as a major impact category.

Regarding the insulation, WGTES has the highest global warming impact (25.6 g $CO_{2\text{-}eq}/kWh_{th}$) due to the extensive use of FGG (837 m³). For the construction of TTES, less FGG was used, resulting in lower impacts (1.18 g $CO_{2\text{-}eq}/kWh_{th}$). In contrast, for the construction of PTES, polymer foam was used (2,422 m³), which has a lower environmental impact (0.42 g $CO_{2\text{-}eq}/kWh_{th}$). Substituting FGG with more environmentally friendly materials could reduce the overall environmental impacts significantly.

Substantial amounts of concrete (898 t) and steel (27 t) were used in the construction of the TTES, leading to overall higher emissions (steel: 2.01 g $CO_{2\text{-}eq}/kWh_{th}$; concrete: 3.3 g $CO_{2\text{-}eq}/kWh_{th}$). In contrast, the underground construction of PTES and WGTES relies on the surrounding soil for structural stability, eliminating the need for additional supporting elements. There, the use of a plastic foil as a sealing material (here: HDPE) is sufficient, leading to lower impacts (PTES: 0.65 g $CO_{2\text{-}eq}/kWh_{th}$). Consequently, fully buried sTES systems should be preferred to reduce the need for static support materials (i.e., steel and concrete) and alternative materials with lower environmental impacts, such as HDPE, should be considered.

### 4.2 EVALUATION WITH OTHER STUDIES

The analysis of the construction phase highlights significant differences in environmental impact among the storage systems (Figure 3).

The WGTES system shows the highest emissions, with 29.4 g $CO_{2\text{-}eq}/kWh_{th}$. The TTES follows with 7.9 g $CO_{2\text{-}eq}/kWh_{th}$, while the ATES (1.9 g $CO_{2\text{-}eq}/kWh_{th}$) (Stemmle et al., 2021) and PTES (1.4 g $CO_{2\text{-}eq}/kWh_{th}$) show the lowest impacts. The differences are attributed to differences in material types, quantities, and/or boundary conditions (e.g., structural requirements). Fully buried systems, such as ATES, WGTES, and PTES, benefit from lower material demands, reducing the associated emissions. In contrast, TTES requires extensive structural reinforcement, leading to higher environmental impacts.

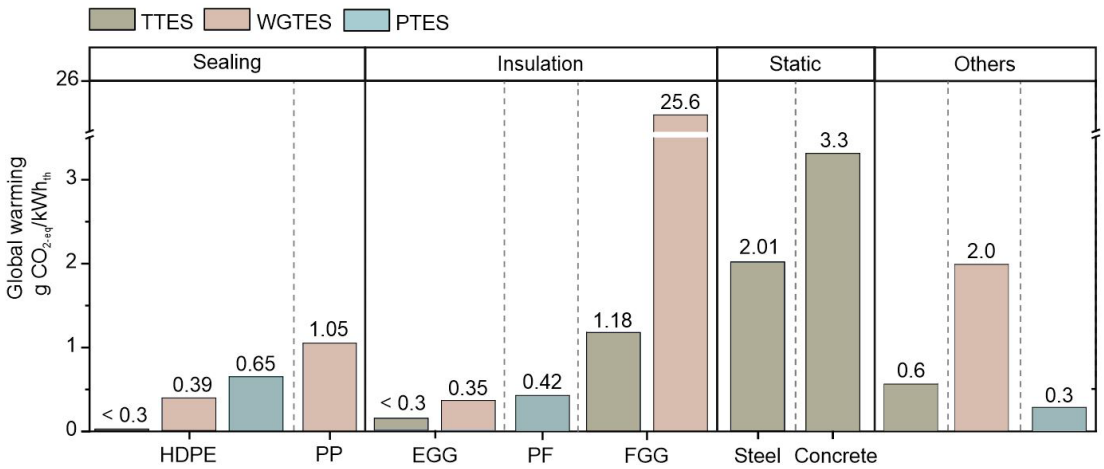


**Figure 2: Material contributions to environmental impact categories of global warming. HDPE: high-density polyethylene, PP: polypropylene, EGG: expanded glass granulate, PF: polymer foam, FGG: foam glass gravel. According to (Weise et al., 2025).**

Comparing these results with the ATES system *Bonner Bogen* (Stemmle et al., 2021) further emphasises the need for optimised material selection And construction strategies to minimise environmental impacts.

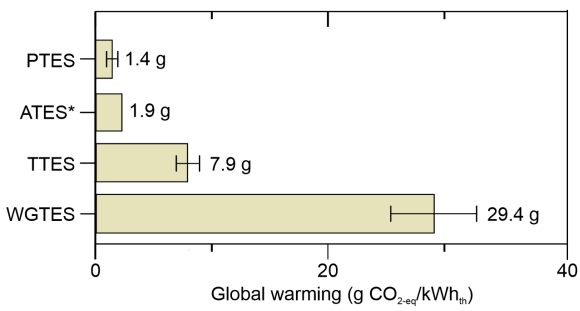


**Figure 3: Evaluation of the global warming (in g $CO_{2\text{-}eq}/kWh_{th}$) of the three selected sTES technologies with the *Bonner Bogen* aquifer thermal energy storage (ATES). * (Stemmle et al., 2021). According to (Weise et al., 2025).**

## 5. CONCLUSION

Seasonal thermal energy storage (sTES) is one key technology to decarbonise modern heat supply systems by conserving surplus summer heat for use in winter. However, the environmental impact of sTES installations has not been well explored so far. For the three sTES variants compared in detail in this study, GHG emissions during the construction phase are between 1.4 and 29.4 g $CO_{2\text{-eq}}/kWh_{th}$. WGTES shows the highest impact in all analysed impact categories, while PTES shows the lowest.

The results underline the need for system-specific environmental optimisation. This can be achieved by reducing the use of building materials such as concrete, steel, and foam glass gravel. The presented comprehensive life cycle assessment already shows that there is potential for improvement in the construction phase of sTES in many respects, although it focuses solely on global warming, whereas additional impact categories such as non-renewable energy and mineral extraction were also considered in Weise et al. (2025).

However, more research is needed to establish sTES as an optimised energy source. It is fundamental to include the operational and end-of-life phases to identify the potential for improvement across the life cycle, and to avoid shifting the environmental burden either between life cycle phases or beyond the system boundaries.

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## REFERENCES

Bayer, P., Saner, D., Bolay, S., Rybach, L., Blum, P., 2012. Greenhouse gas emission savings of ground source heat pump systems in Europe: A review. Renewable and Sustainable Energy Reviews 16 (2), 1256–1267.

Bloemendal, M., Olsthoorn, T., Boons, F., 2014. How to achieve optimal and sustainable use of the subsurface for Aquifer Thermal Energy Storage. Energy Policy 66, 104–114.

Bodmann, M., Mangold, D., Nußbicker, J., Raab, S., Schenke, A., Schmidt, T., 2005. Solar unterstützte Nahwärme und Langzeit-Wärmespeicher.

Bott, C., Dressel, I., Bayer, P., 2019. State-of-technology review of water-based closed seasonal thermal energy storage systems. Renewable and Sustainable Energy Reviews 113, 1–16.

Dannemand Andersen, J., Bødker, L., Jensen, M.V., 2013. Large Thermal Energy Storage at Marstal District Heating. Proceedings of the 18th International Conference on Soil Mechanics and Geotechnical Engineering, 3351-3354.

DIN EN ISO 14040, 2021. Umweltmanagemt - Ökobilanz - Grundsätze und Rahmenbedingungen +. Beuth Verlag GmbH, Berlin 13.020.10. doi:10.31030/3179655, 36 pp.

DIN EN ISO 14044, 2018. Umweltmanagement - Ökobilanz - Anforderungen und Anleitungen. Beuth Verlag GmbH, Berlin 13.020.10. doi:10.31030/3179656. Accessed 25 February 2024, 65 pp.

Fleuchaus, P., Godschalk, B., Stober, I., Blum, P., 2018. Worldwide application of aquifer thermal energy storage – A review. Renewable and Sustainable Energy Reviews 94, 861–876.

Guadalfajara, M., Lozano, M.A., Serra, L.M., 2014. Analysis of Large Thermal Energy Storage for Solar District Heating, 11 pp. Accessed 24 February 2024.

Holm, L., PlanEnergi, 2021. Entwicklung von großen thermischen Energiespeichern in Dänemark.

Jensen, M.V., PlanEnergi, 2014. Seasonal pit heat storages - Guidelines for materials & construction. SHC.

Jolliet, O., Margni, M., Charles, R., Humbert, S., Payet, J., Rebitzer, G., Rosenbaum, R., 2003. IMPACT 2002+: A new life cycle impact assessment methodology. Int J LCA (The International Journal of Life Cycle Assessment) 8 (6), 324–330.

Keil, C., Plura, S., Radspieler, M., Schweigler, C., 2008. Application of customized absorption heat pumps for utilization of low-grade heat sources. Applied Thermal Engineering 28 (16), 2070–2076.

Mangold, D., Miedaner, O., Tziggili, E.P., Schmidt, T., Unterberger, M., Zeh, B., 2012. Technisch-wirtschftliche Analyse und Weiterentwicklung der solaren Langzeit-Wärmespeicherung: Forschungsbericht zum BMU-Vorhaben 0329607N. Wissenschaftlich-technische Programmbegleitung für Solarthermie2000Plus. Steinbeis Forschungsinstitut für solare und zukunftsfähige thermische Energiesysteme.

Mangold, D., Riegger, M., Schmidt, T., 2007. Solare Nahwärme und Langzeit-Wärmespeicher: Wissenschaftlich-technische Programmbegleitung für Solarthermie2000+.

Mangold, D., Schmidt, T., Dohna, A., Späh, D., 2016. Guideline for Seasonal Thermal Energy Storage Systems in the Built Environment. Solites, Steinbeis Research Institute for Solar and Sustainable Thermal Energy Systems, 1–62.

Marx, R., Bauer, D., Drück, H., 2012. Schüttfähige Dämmstoffe für Langzeit-Wärmespeicher: Erfahrungen und Planungsempfehlungen.

Ni, Z., Wang, Y., Wang, Y., Chen, S., Xie, M., Grotenhuis, T., Qiu, R., 2020. Comparative Life-Cycle Assessment of Aquifer Thermal Energy Storage Integrated with in Situ Bioremediation of Chlorinated Volatile Organic Compounds. Environmental science & technology 54 (5), 3039–3049.

Ochs, F., Nußbicker-Lux, J., Marx, R., Koch, H., Heidemann, W., Müller-Steinhagen, H., 2008. Solar assisted district heating system with seasonal thermal energy storage in Eggenstein-Leopoldshafen.

Pinel, P., Cruickshank, C.A., Beausoleil-Morrison, I., Wills, A., 2011. A review of available methods for seasonal storage of solar thermal energy in residential applications. Renewable and Sustainable Energy Reviews 15 (7), 3341–3359.

Reuß, M., 2016. Solar District Heating - an Innovative Approach of an Established Technology.

Röpke, I., 2007. Sonnensaison bis in den tiefen Winter. Sonne, Wind und Wärme (10), 46–52.

Schneider, B., 2009. Schulsanierung mit solarer Nahwärme verknüpft. Bine Informationsdienst (4).

Schüppler, S., Fleuchaus, P., Blum, P., 2019. Techno-economic and environmental analysis of an Aquifer Thermal Energy Storage (ATES) in Germany. Geotherm Energy 7 (1).

Sørensen, P.A., Schmidt, T., 2018. Design and Construction of Large Scale Heat Storages for District Heating in Denmark. EnerSTOCK2018.

Spliethoff, H., Stimming, U., 2010. Begleitforschung Solare Nahwärme Am Ackermannbogen in München - SNAB. ZAE Bayern.

Stemmle, R., Blum, P., Schüppler, S., Fleuchaus, P., Limoges, M., Bayer, P., Menberg, K., 2021. Environmental impacts of aquifer thermal energy storage (ATES). Renewable and Sustainable Energy Reviews 151, 111560.

Tomasetta, C., van Ree, C.C.D.F., Giffioen, J. (Eds.), 2015. Life Cycle Analysis of Underground Thermal Energy Storage. Springer International Publishing, Schweiz.

Weise, J., Bott, C., Menberg, K., Bayer P., 2025. Comprehensive life cycle assessment of selected seasonal thermal energy storage systems, 2025, submitted.

Xiang, Y., Xie, Z., Furbo, S., Wang, D., Gao, M., Fan, J., 2022. A comprehensive review on pit thermal energy storage: Technical elements, numerical approaches and recent applications. Journal of Energy Storage 55, 105716.

Xu, J., Wang, R.Z., Li, Y., 2014. A review of available technologies for seasonal thermal energy storage. Solar Energy 103, 610–638.